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NCFE · LEVEL 2 · MATHEMATICS

NCFE Functional Skills Mathematics Level 2

2023 Section B · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: NCFE-L2-Maths-Practice-Paper-Section-B

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

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Activity 2: Hotel

2 (a) Fiona is on a work placement in a hotel.

The hotel has a loyalty programme for the guests.

There are three million, four hundred thousand and nineteen guests currently signed up to the loyalty programme worldwide.

Q2(a) Worked Solution:

Write three million, four hundred thousand and nineteen as a number.

[1 mark]

3 400 019

[1 mark]

Your answer:

2 (b) There are 216 guests currently staying at the hotel.

81 of these guests are signed up to the loyalty programme.

Fiona thinks that more than $\frac{4}{10}$ of these guests are signed up to the loyalty programme.

Is she correct?

Show how you decide.

[2 marks]

Q2(b) Worked Solution:

See below

1

81

216

Allow 216 × 0.4 or 86.4

1

No and

1S

40 and

16

[2 marks]

Your answer:

Please turn over

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2 (c) The table below shows information about the 216 guests staying at the hotel on Sunday.

Sunday	
Number of nights stay	**Number of guests**
1	59
2	42
3	65
4	25
5	16
6	5
7	4
Total	216

Every Sunday and Monday, guests who will be staying for more nights than the mode are given a voucher for £10 off food and drink in the restaurant.

Q2(c) Worked Solution:

(£) 520

Award 4 marks if correct answer given

1

3

May be implied by 25 + 16 + 5 + 4 or 50

H23b

1

(25 + 16 + 5 + 4) × 10 or 500

OR

25 + 16 + 5 + 4 or 50

[4 marks]

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The total value of the vouchers given to guests on Monday was 4% more than the total value of the vouchers given on Sunday.

What was the total value of the vouchers given on Monday?

[4 marks]

Your answer: £

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2 (d) Fiona uses this formula to work out the number of loyalty points earned by each guest last year.

$$L = 300\ (s + \frac{r}{2})$$

Where L = number of loyalty points collected in a year
s = number of nights stayed in a suite
r = number of nights stayed in a regular room

She then uses this table to put each guest into one of five tiers based on the number of loyalty points they earned.

Q2(d) Worked Solution:

See below

1

Tier	Loyalty points collected in a year
1	0 – 599
2	600 – 1499
3	1500 – 2999
4	3000 – 4999
5	5000+

300 × (3 + 7 ÷ 2) or 1950

1

Last year, Mrs Jones stayed 3 nights in a suite and 7 nights in a regular room.

Which tier should Fiona put Mrs Jones into?

You must show all your working.

[2 marks]

(Tier) 3 and 1950

OE tier 3 supported by correct working

[2 marks]

Your answer: **Tier**

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Examiner use only

2 (e) A luxury room in the hotel has a bathtub 60.3 inches long.

What is the length of the bathtub in centimetres?

Use the conversion 1 inch = 2.54 cm

Give your answer correct to 1 decimal place.

[2 marks]

Q2(e) Worked Solution:

153.2(cm)

Award 2 marks if correct answer given

1

60.3 × 2.54 or 153.162

1

153.2(cm)

[2 marks]

Your answer: **cm**

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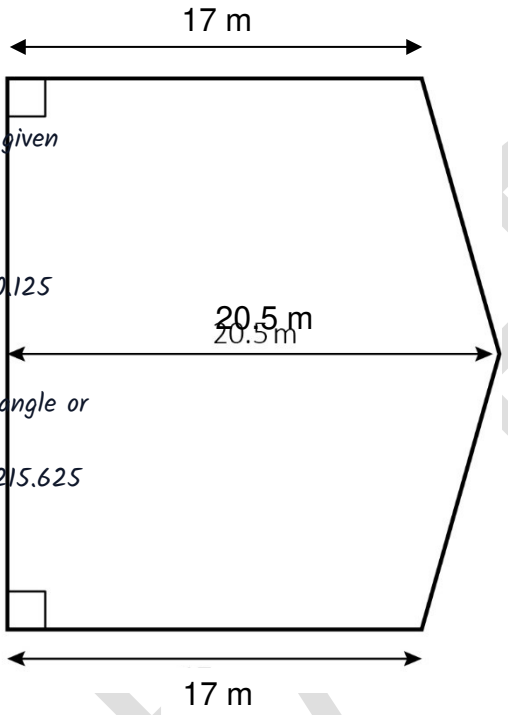
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Examiner use only

2 (f) Fiona helps with organising an event at the hotel.

The diagram below shows the dimensions of the dance floor.

The shape of the dance floor has one line of symmetry.



Not drawn accurately

Q2(f) Worked Solution:

479

Award 4 marks if correct answer given

1

0.5 x (20.5 -17) x 23 or 40.25

OR

0.5 x (20.5 -17) x 0.5 x 23 or 20.125

OE correct method for area of triangle or trapezium e.g.

(20.5 + 17) ÷ 2 × (23 ÷ 2) or 215.625

1

[4 marks]

On the dance floor, each person must have at least 0.9 m^2 of floor space.

Work out the maximum number of people who can use this dance floor at one time.

[4 marks]

Your answer: **people**

[Total marks: 15]

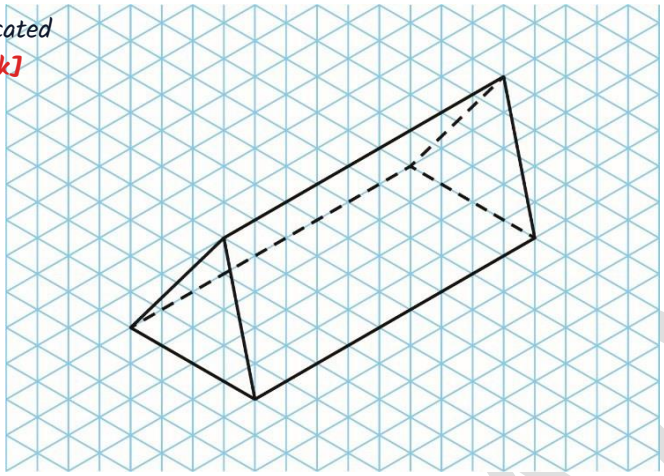
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Activity 3: Product design

3 (a) Calista is a product designer at a toy factory.

The diagram shows a metal part in a toy robot.

Q3(a) Worked Solution:
C indicated
[1 mark]



What is the name of this shape?

Tick the correct answer.

- ☐ A pyramid
- ☐ B cylinder
- ☐ C prism
- ☐ D cuboid

[1 mark]

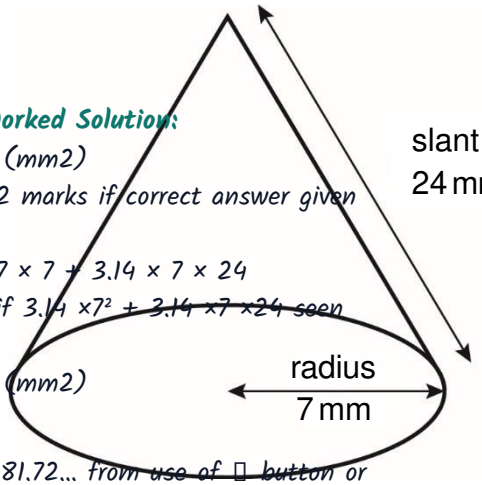
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Examiner use only

3 (b) Calista designs a plastic piece for a game.

The piece is in the shape of a cone.
The diagram shows the dimensions of the piece.



Not drawn accurately

Q3(b) Worked Solution:
681.38 (mm2)
Award 2 marks if correct answer given
1
3.14 × 7 × 7 + 3.14 × 7 × 24
Award if 3.14 ×7² + 3.14 ×7 ×24 seen
1
681.38 (mm2)

Allow 681.72... from use of π button or 681.4

Work out the surface area of the piece.

Use $A = \pi r^2 + \pi rs$
where A = surface area of a cone
r = radius of the circular base
s = slant height of the cone

Use $\pi = 3.14$

[2 marks]

Implies first mark
[2 marks]

Your answer: **mm²**

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Examiner use only

3 (c) Calista mixes three colours to make teal-coloured paint for the plastic game pieces.
She mixes blue, green and yellow paint in the ratio 4 : 2 : 1

Calista wants to make 14 litres of teal-coloured paint.
She thinks she needs 1.6 litres of yellow paint.

Q3(c) Worked Solution:

Is she correct?

Show how you decide.

[2 marks]

No AND 2 (litres)
OR
No AND 11.2 (litres)
Award 2 marks if correct answer given
1
14 ÷ (4 + 2 + 1) or 2
OR
1.6 × (4 + 2 + 1) or 11.2
7
Level 2 – P001457-PB P10 v1.5
[2 marks]

Your answer:

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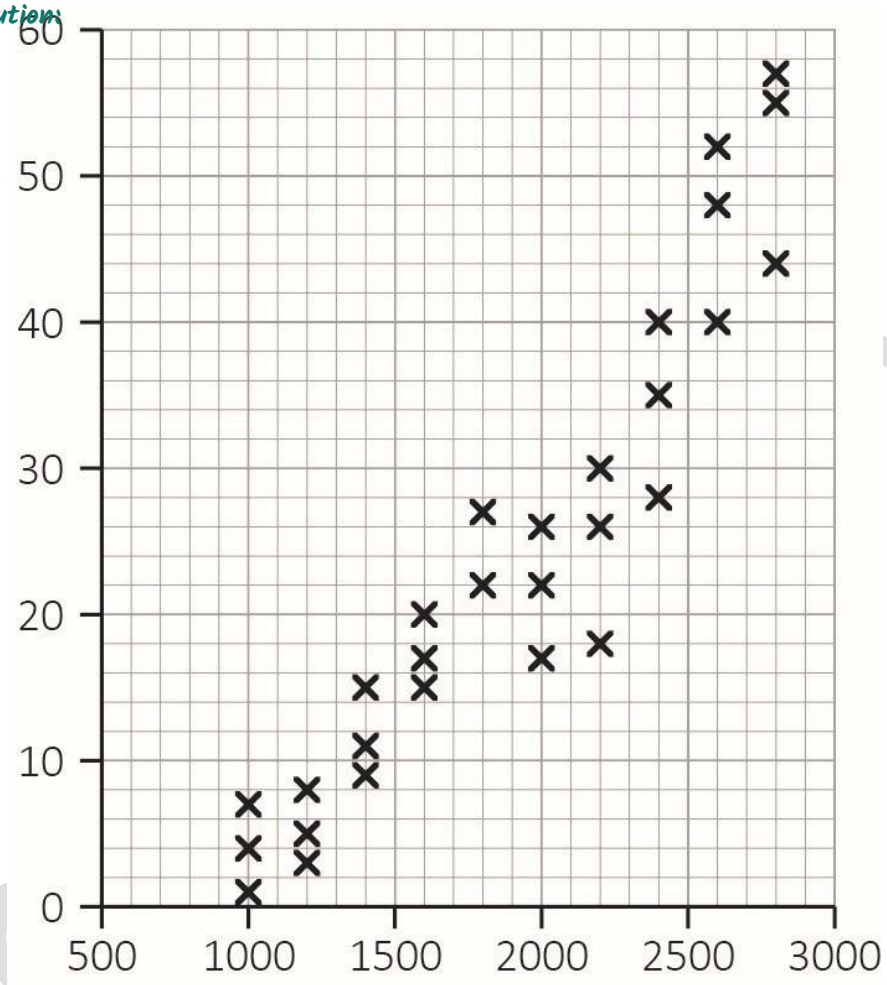
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Examiner use only

3 (d) The graph below shows the information about the settings and fault rate of a machine in the toy factory.

Machine setting and fault rate

Q3(d) Worked Solution:
20
H28
[1 mark]



What was the highest fault rate when the machine was set to make 1600 toys per hour?

[1 mark]

Your answer:	**faulty toys per 1000**

Examiner use only

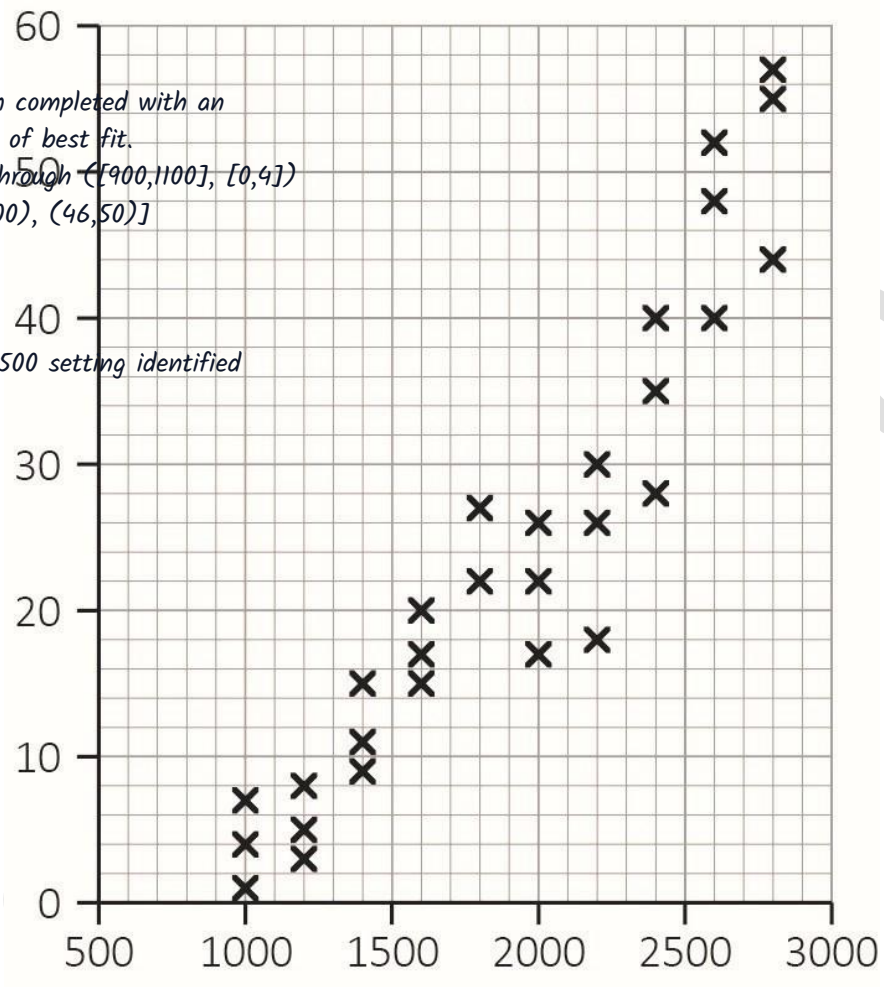
3 (e) The graph below shows the information about the settings and fault rate of a machine in the toy factory.

Machine setting and fault rate

Q3(e) Worked Solution:

See below
I
Scatter diagram completed with an appropriate line of best fit.
Line must go through ([900,1100], [0,4]) and ([2700,2900), (46,50)]
H28
I
[38, 42]
Fault rate at 2500 setting identified

[5 marks]



Examiner use only

Calista is writing a report.
She uses a line of best fit to find the fault rate when the machine setting is 2500

Calista writes, “The fault rate when the machine setting is 2500 as a percentage of the median fault rate when the machine setting is 2600 is more than 90%.”

Is she correct?

Show how you decide.

[5 marks]

Your answer:	

PAST PAPER

Please turn over

Examiner use only

3 (f) Calista has this data about the volume of plastic used to make the toys she has designed.

Volume of plastic (p in cm^3) used to make a toy	Number of toys made
$0 < p \leq 30$	26 000
$30 < p \leq 60$	42 000
$60 < p \leq 90$	17 000
$90 < p \leq 120$	9000
$120 < p \leq 150$	6000
Total:	100 000

Q3(f) Worked Solution:
48.9
Award 4 marks if the answer is seen
1

She works out an estimate of the mean volume of plastic used for each toy.

She wants to reduce this mean by 8%

If she is successful, what will the new mean be?

Give your answer correct to 1 decimal place.

[4 marks]

(26000 × 15) + (42000 × 45) + (17000 × 75) + (9000 × 105) + (6000 × 135) or 5 310 000
Must use midpoints
Allow not using 000 in each figure
H24
1
[4 marks]

Your answer:

[Total marks: 15]

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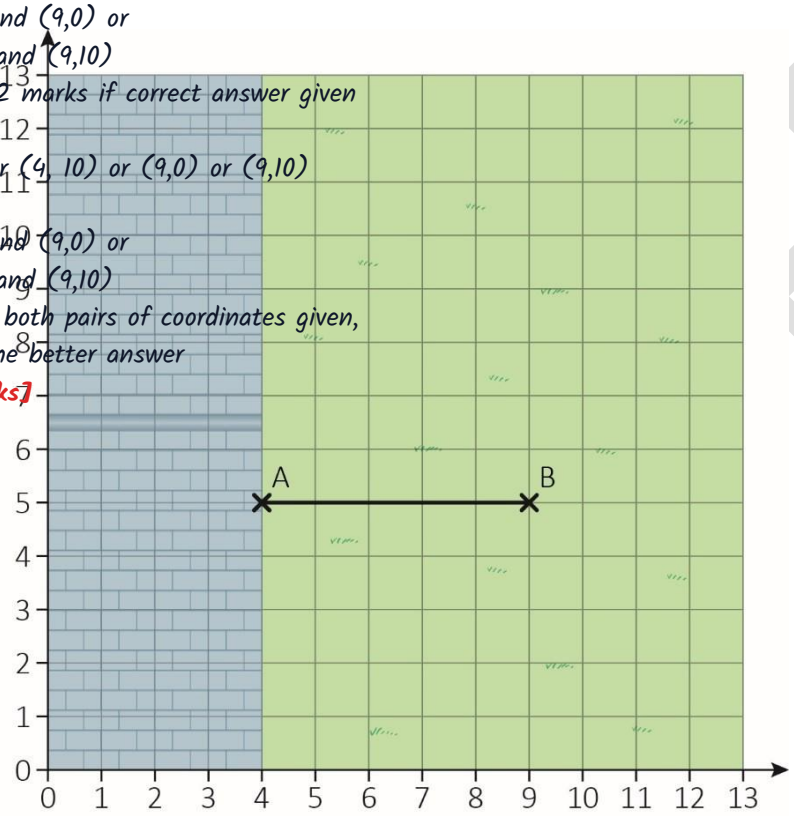
Examiner use only

Activity 4: House extension

4 (a) Abdul wants to build an extension to his house.

The coordinate grid below shows the plan of his back garden where the extension will be.

Q4(a) Worked Solution:
(4,0) and (9,0) or
(4,10) and (9,10)
Award 2 marks if correct answer given
I
(4,0) or (4, 10) or (9,0) or (9,10)
I
(4,0) and (9,0) or
(4,10) and (9,10)
CAO If both pairs of coordinates given,
mark the better answer
[2 marks]



Points A and B are two corners of the extension.
The line AB shows one wall of the extension.

The floor of the extension will be a square.

Write down a set of possible coordinates for the other two corners of the extension.

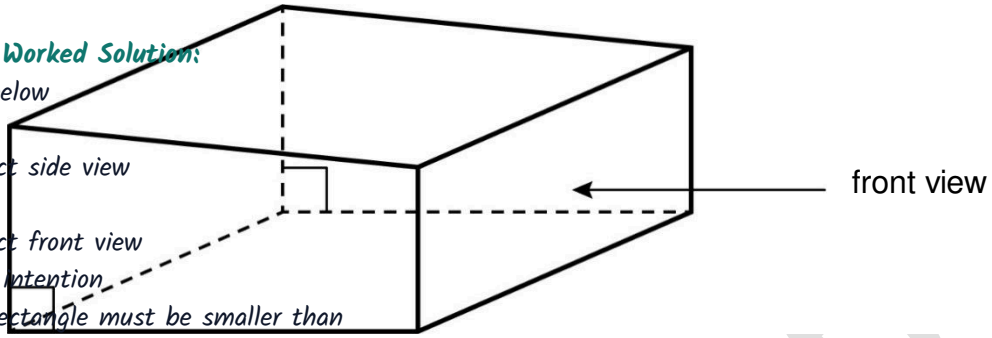
[2 marks]

Your answer:

Please turn over

Examiner use only

4 (b) The diagram shows the shape of the extension.



Q4(b) Worked Solution:
See below
1
Correct side view
1
Correct front view
Mark intention
Top rectangle must be smaller than bottom
[2 marks]

Draw the side view **and** the front view of the extension.

[2 marks]

Side view
Front view

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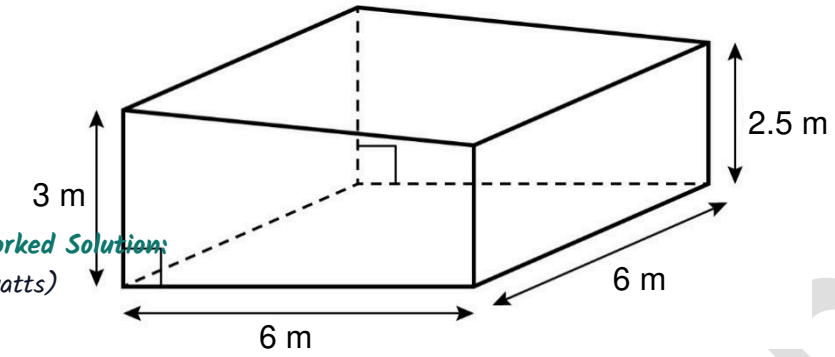
Please turn over for the next question.

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Examiner use only

4 (c) The diagram below shows the dimensions of the extension.

Not drawn accurately



Q4(c) Worked Solution:
4160 (watts)

Award 4 marks if correct answer given

1 Abdul wants to work out the power of the heater needed for the extension.

First, he needs to find the volume (V) of air inside the empty extension by using this formula:

$V = 0.5 \times w \times l \times (a + b)$

where a = greatest height (m)
b = smallest height (m)
w = width (m)
l = length (m)

0.5 × (3 + 2.5) × 6 × 6
OE full correct method for total volume,
e.g.
6 × 2.5 × 6 + (3 − 2.5) × 6 × 0.5 × 6 or
6 × 3 × 6 − (3 − 2.5) × 6 × 0.5 × 6
1
99
[4 marks]

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Abdul then uses this formula to work out the power of the heater needed for the extension.

$P = 40V + 100n$

where P = power of the heater (watts)
V = volume of air inside the empty extension (m^3)
n = the number of windows

There will be 2 windows.

Work out the power of the heater needed for Abdul's extension.

[4 marks]

Your answer: **watts**

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Please turn over

Examiner use only

4 (d) Abdul needs to make a planning application.
He must include a scale drawing of the garden and the extension.

The width of the garden is 11.8 metres.
The drawing will use the scale 1 : 200

Abdul thinks that the width of the garden in the scale drawing is 5.4 cm long.

Q4(d) Worked Solution:

Is Abdul correct?

Show how you decide.

[2 marks]

No and 5.9 (cm) or
No and 10.8 (m)
Award 2 marks if correct answer given
1
11.8 × 100 ÷ 200 or 5.9 or
5.4 × 200 ÷ 100 or 10.8
OE any full correct method
1
No and 5.9 (cm) or
No and 10.8 (m)
[2 marks]

Your answer:	

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Examiner use only

4 (e) Abdul finds this information about planning applications to his local council in 2020

Type of planning application	Received	Granted	Declined
Major	270	195	75
Residential	306	267	39
Commercial	124	94	30

Q4(e) Worked Solution:

Abdul thinks that, overall, residential applications are more likely to be granted than commercial applications.

Is he correct?

Yes AND 0.8(7...) and 0.7(5...)

Show how you decide.

[2 marks]

Award 2 marks if correct answer given
1
267
306 or
94
124
OE any full correct method
H27
9
[2 marks]

Your answer:

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4 (f) Abdul finds this information about planning applications to his local council in 2020

Type of planning application	Received	Granted	Declined
Major	270	195	75
Residential	306	267	39
Commercial	124	94	30

Q4(f) Worked Solution:

30

What is the probability that one of these applications chosen at random is a declined commercial application?

[1 mark]

700 or 0.04(2...) or 3/70 or 4.2%

OE allow any correct rounding

Award if fraction stated as 3 out of 7, 3 over 7, etc

H26

[1 mark]

Your answer:

4 (g) Abdul is granted planning permission.

He does this calculation to work out how much concrete to buy:

$$\frac{(0.2 + 0.05) \times 6^2}{0.9}$$

Q4(g) Worked Solution:

10

What answer should Abdul get?

[2 marks]

Award 2 marks if correct answer given

1

9

1

10

[2 marks]

Your answer:

[Total marks: 15]

This is the end of the assessment.

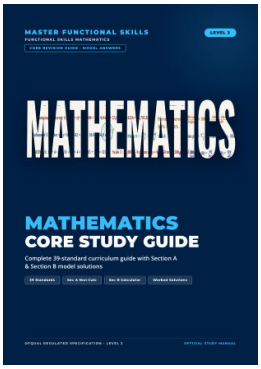
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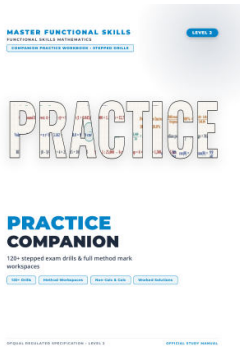


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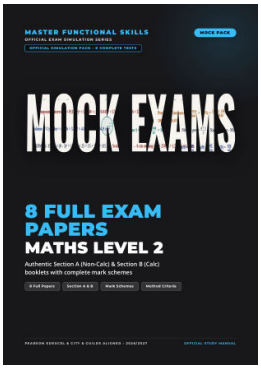


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